PROJECT PROPOSAL

NurtureShield

Machine Learning-Based Malicious URL and Phishing Detection System

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1. Project Overview

NurtureShield is a web-security prototype for detecting phishing and malicious URLs before a user opens them. The system analyzes lexical and structural properties of a submitted URL and uses a supervised machine learning classifier to estimate whether the link is legitimate or potentially malicious. Rather than returning only a label, the prototype will also explain the main URL characteristics that influenced the assessment. The final demonstration will include a browser extension that can assess links before navigation, providing a more practical security workflow than a standalone URL textbox.

2. Problem Statement

Phishing remains effective because malicious links can closely resemble legitimate websites. Attackers use misleading domains, deceptive subdomains, IP-based addresses, Unicode lookalikes, unusual encoding, suspicious keywords, shortening services, and complex URL structures to conceal a destination. Blacklists can block previously identified threats but may not immediately protect against newly created or modified links. NurtureShield therefore focuses on URL-level risk estimation that can complement blacklist-based defenses without intentionally executing potentially harmful webpage content.

3. Project Objectives

- Develop a working prototype that detects phishing and malicious URL patterns.
- Prepare and preprocess a labeled dataset of legitimate and malicious/phishing URLs while explicitly preventing train-test leakage.
- Extract lexical and structural security features, including domain, subdomain, encoding, Unicode, IP-address, length, digit, special-character, and suspicious-token indicators.
- Train and compare suitable machine learning classifiers against a simple non-ML baseline to demonstrate whether machine learning adds measurable value.
- Provide an explanation with each prediction by highlighting suspicious URL characteristics or influential features.
- Test robustness against realistic URL variations such as encoding changes, misleading subdomains, Unicode domains, and other transformations.
- Build a browser extension that can assess a link before navigation and present a clear risk status and warning.
- Evaluate performance using precision, recall, F1-score, confusion matrix, false-positive/false-negative analysis, generalization testing, and robustness results.

4. Proposed Solution and Methodology

The project will use supervised machine learning for URL classification. A labeled dataset will first be cleaned, deduplicated, and inspected for repeated or near-duplicate URLs. Data splitting will be designed to reduce leakage, with related URLs/domains kept from being unintentionally distributed across training and test sets where this could inflate performance. A held-out test set will remain unseen until final evaluation.

Each URL will be transformed into security-related lexical and structural features obtainable without opening the target page. Candidate features include URL length, domain and subdomain structure, IP-address usage, digits and special characters, suspicious terms, HTTPS indicators, encoding patterns, Unicode/Punycode signals, path/query

complexity, and related structural properties. A simple rule-based or statistical baseline will be evaluated alongside machine learning classifiers.

After model selection, the same preprocessing and feature-extraction pipeline will be integrated into the application and browser extension. For every prediction, the interface will show the risk result together with interpretable indicators such as an IP-based host, excessive subdomains, suspicious tokens, unusual encoding, or other influential features. The browser extension will allow a user to assess a link before navigation rather than requiring manual copying into a textbox.

Link / URL	Feature Extraction	Baseline + ML Models	Risk Assessment	Prediction Explanation	Browser Warning

5. Project Scope

The scope covers dataset preparation, leakage-resistant splitting, URL feature extraction, baseline and machine learning model development, prediction explanations, robustness testing, application integration, and a browser-extension demonstration. The system primarily relies on information derivable from the URL itself, allowing analysis without intentionally loading suspicious webpage content.

The project will not operate as a complete antivirus suite, remove malware, perform penetration testing, execute downloaded malware, or conduct full dynamic webpage analysis. These activities remain outside the intended URL-detection problem.

6. Significance and Security Applications

NurtureShield provides an additional security layer at the point where a user is about to interact with an unknown link. Its browser-extension demonstration makes the project directly applicable to pre-navigation link checking, while the explanation component makes decisions easier to understand and defend. The underlying approach can also inform email or messaging filters, organizational security gateways, and security-awareness tools.

7. Evaluation Plan

Evaluation will emphasize credible generalization rather than accuracy alone. The final test set will be isolated from model development, and duplicate or strongly related samples will be controlled to reduce leakage. Machine learning models will be compared with a simple baseline so that improvement can be quantified.

- Report accuracy, precision, recall, F1-score, and confusion matrix, with particular attention to malicious-URL recall and false negatives.
- Perform error analysis on false positives and false negatives and identify feature patterns associated with mistakes.
- Assess generalization on unseen data and, where feasible, domain-aware or source-aware splits rather than relying only on a random split.
- Run robustness tests using controlled URL variations such as percent/character encoding, misleading subdomains, Unicode/Punycode domains, added path/query noise, and similar transformations.
- Report baseline-versus-ML results and discuss limitations of URL-only detection and cases requiring additional security checks.

8. Planned Deliverables

- Complete source code for the NurtureShield prototype and browser extension.
- Cleaned dataset preparation, leakage-resistant split logic, and reproducible preprocessing/feature-extraction pipeline.
- Simple baseline plus trained machine learning model(s) and comparative results.
- Working web interface and browser extension for pre-navigation URL assessment.
- Per-prediction explanations showing relevant suspicious indicators or influential features.
- Robustness-test suite for representative URL variations.
- Installation/run instructions, architecture documentation, and a security analysis covering results, limitations, and lessons learned.